

**NEBULAE**<sup>®</sup>  
DESIGN TO CONNECT

**CONNECT**  
ANYTHING WITH  
EVERYTHING

A Business Vertical Of



Designing Through Innovation

# SYSTEM LEVEL SOLUTIONS

## PRODUCT & SERVICE COMPANY

- No. of business verticals: 6+
- Operating Base: India, UK, USA
- In-house Design & Manufacturing Capability
- 300+ In house staff member

### QUICK FACTS

**100+**

Active projects  
across the globe

Part of



Initiative

**20**

Years and  
counting

# ABOUT

Nebulae, an agile and futuristic IoT platform which enables the integration of the hardwares to a centralised platform, driving interoperability and inter-portability to provide a seamless & compatible connection of devices across Multi-band networks and mediums.



# How Nebulae is an **ENABLER?**



It offers a **Common interface** for devices across all levels



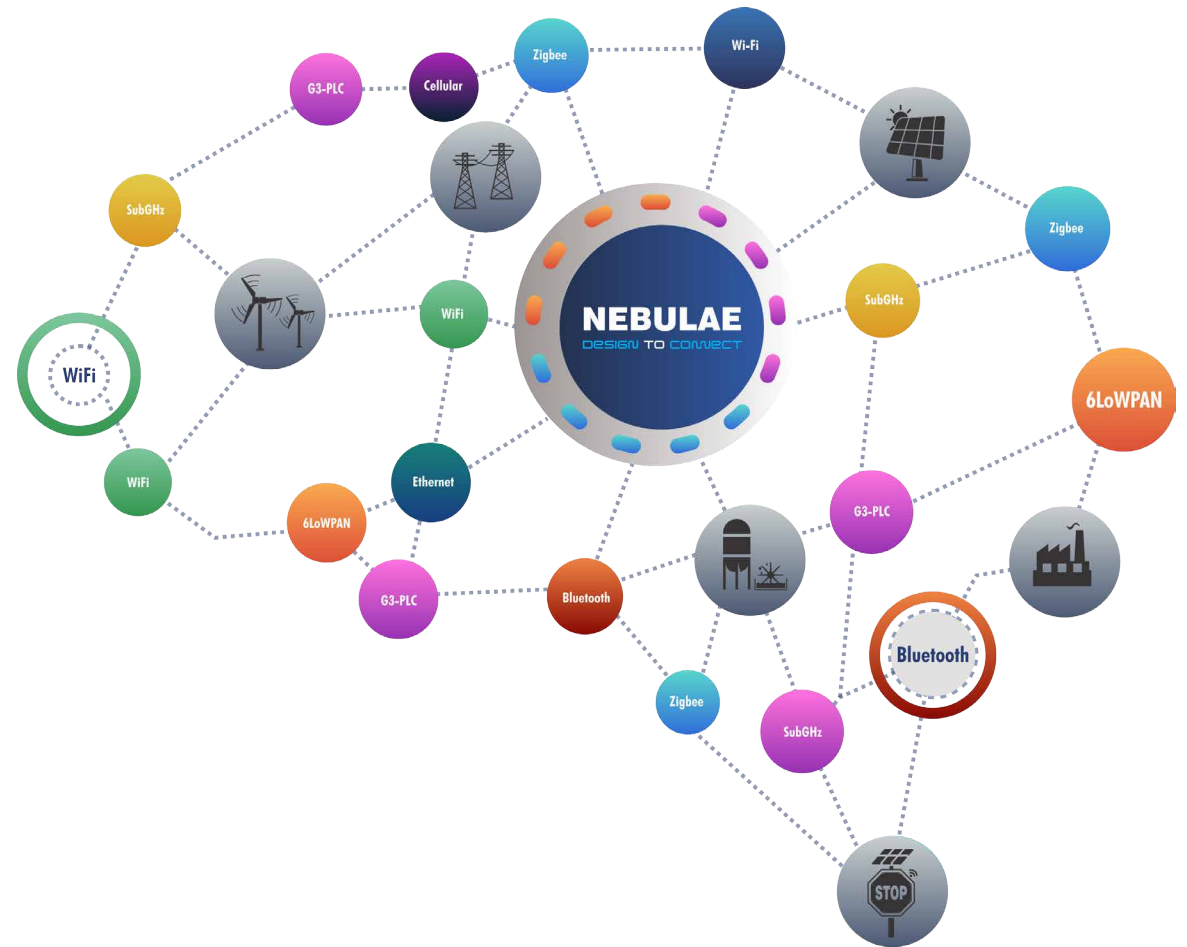
It can communicate across **different communication mediums**



It embeds **Advanced Middleware** as a core of the IoT solution



Underpinned by **Advanced Security and FOTA management** to ensure safety

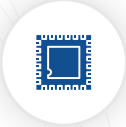


# FEATURES



## ENTERPRISE-GRADE SECURITY

Nebulae's holistic approach provides the full IoT spectrum, from a connected hardware to the cloud



## HARDWARE AGNOSTIC

Nebulae works with multiple hardware platforms making integration with cloud easier



## ZERO CONFIGURATION

NebuLink Middleware (Cloud) services provide zero configuration connection of sensors, devices, and equipment



## EASY AND FAST DEVELOPMENT

Nebulae SDKs speed up the development of the cloud and device applications resulting in fast product development



## SCALABILITY

Nebulae delivers scalability in the devices, cloud and software application with better security, reliability, reduced latency and better user experience

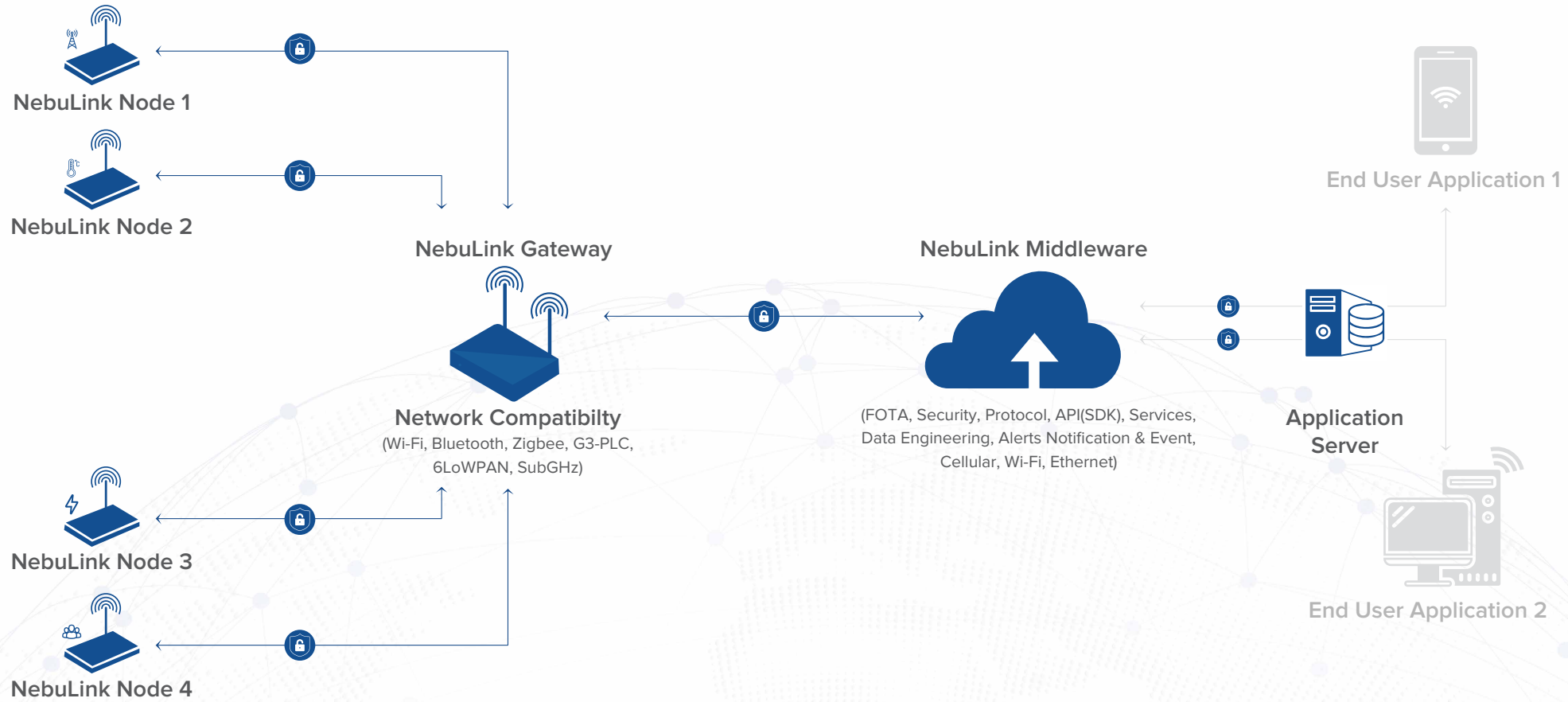


## NETWORK SERVICE

Nebulae's intelligent, self-healing, and self-configuring network service provides a guaranteed data delivery



# ECO-SYSTEM



# NEBULAE PRODUCTS

- NEBULINK NODE
- NEBULINK MIDDLEWARE
- NEBULINK GATEWAY
- NEBULINK FOTA

# NEBULINK NODE

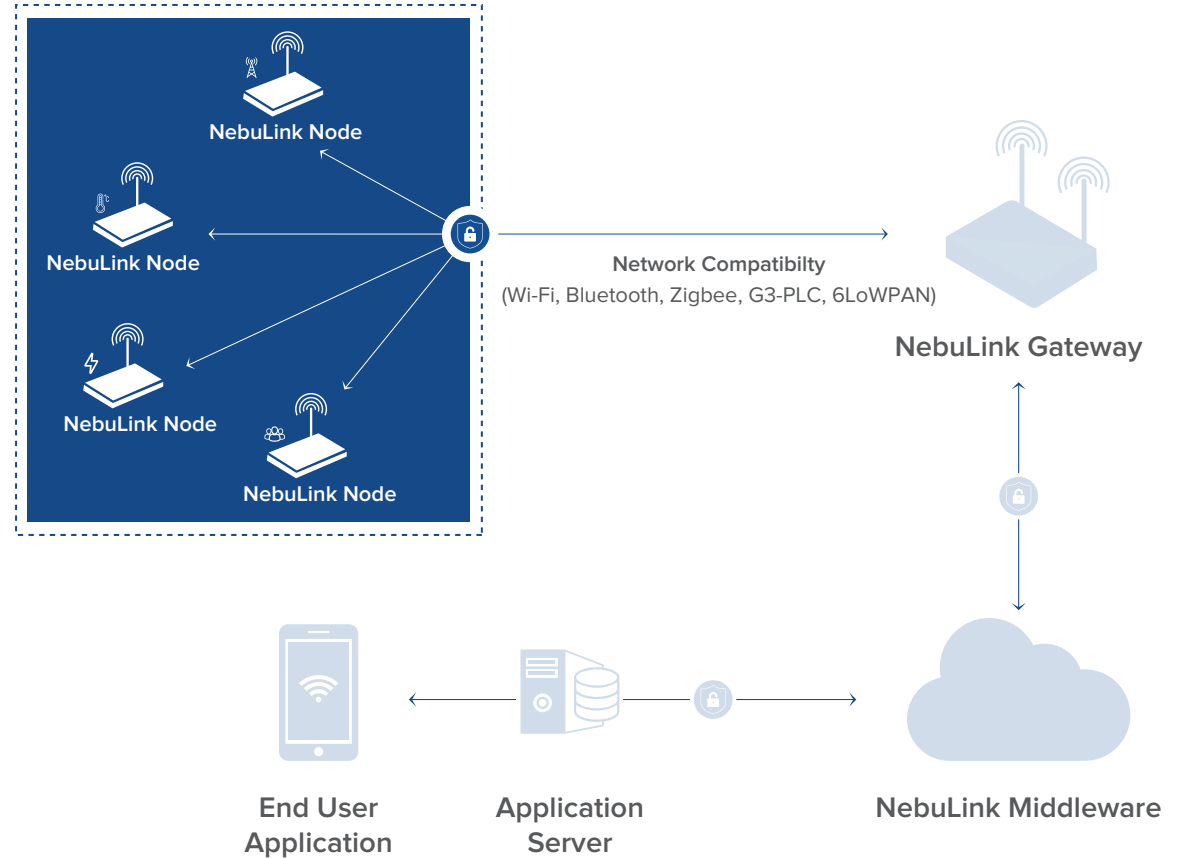
NebuLink Node, a broad-level component of Nebulae **provides wired and wireless connectivity** for sensors, devices and other equipment.

## Features :

- A feature-rich IoT SDK
- Mesh Network
- Rich interface for IO control
- Scalability
- Recovers data against runtime failure

## EVALUATION BOARD

Evaluation Board provides an **easy interface to validate a specific use case** using NebuLink Node SDKs.





# NEBULINK NODE

## 2.4 GHz •

NLN100a, NLN100b  
NLN100c, NLN100d

## SubGHz •

NLN100e

## GSM •

NLN600b (2G)  
NLN600c (4G)

## NB-IoT •

NLN600d

## G3-PLC •

NLN300

## TYPES & APPLICATION

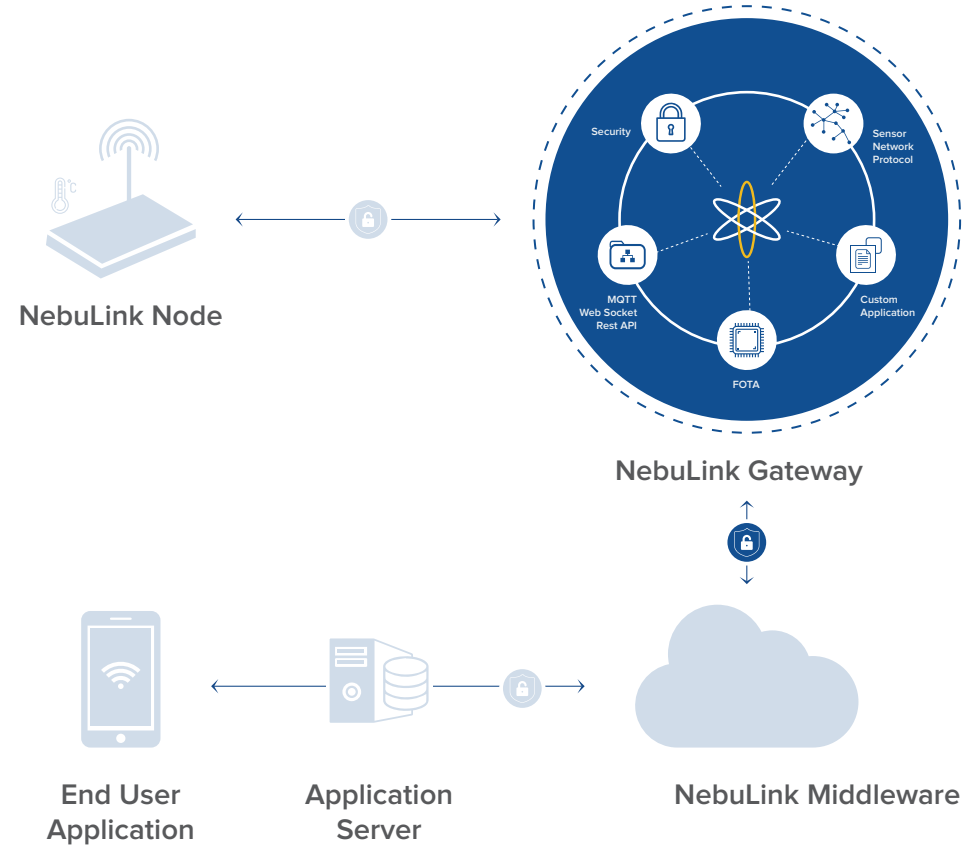
**S** Grid  
**M** Health  
**A** Building  
**R** Parking  
**T** Agriculture  
Manufacturing

# NEBULINK GATEWAY

NebuLink Gateway offers various key ingredients to **access a vast range of IoT functionality** for various applications.

## Features :

- A feature-rich IoT SDK based on FreeRTOS and LINUX
- Cross-industry protocol
- Local event logging capability
- Zero Configuration
- Integration of network technologies
- Remote IO control
- Supports from the smallest 32-bit micro-controller to FPGA-SoC's
- In-build crypto security



# NEBULINK GATEWAY

## Commercial Gateways •

- | NLG1100
- | NLG1200
- | NLG1300

## Industrial Gateways •

- | NLG2200
- | NLG5100
- | NLG5700
- | NLG6700
- | NLG7600

## TYPES & APPLICATION



Real-time data acquisition, processing and transmission



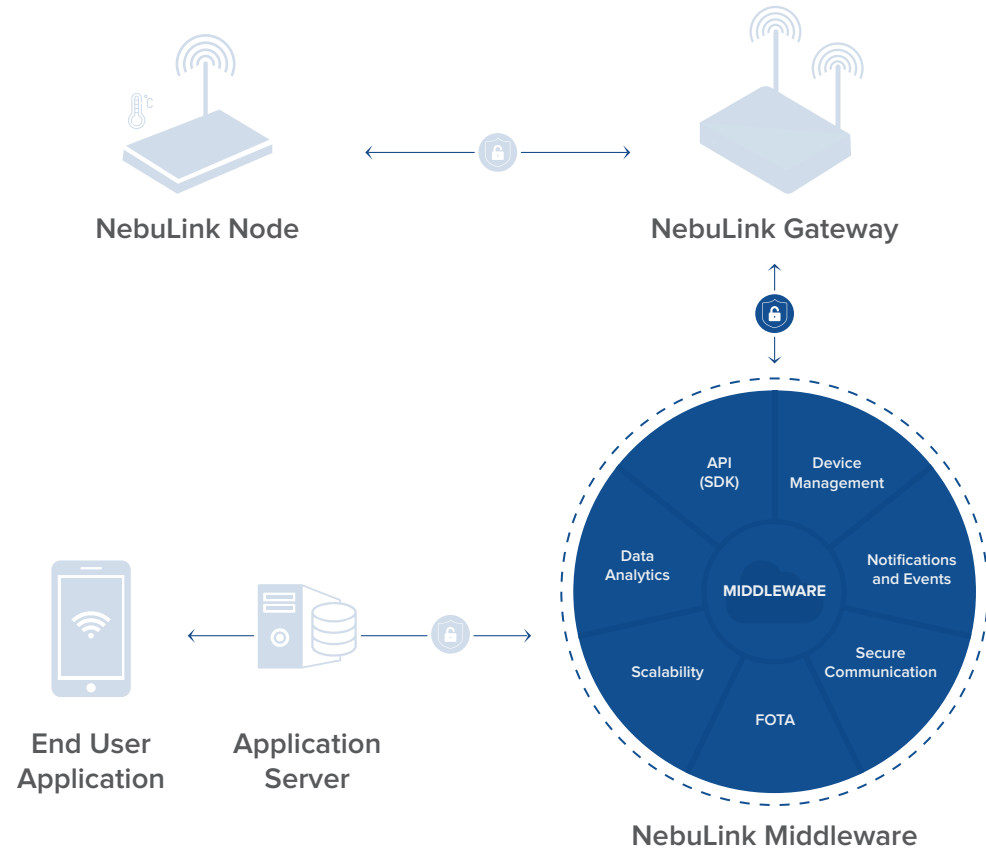
Remote management of devices and assets

# NEBULINK MIDDLEWARE

NebuLink Middleware, a **mediator between the hardware and application**. It integrates nodes and gateways with the applications using SDKs thus allows to connect, operate and automate IoT devices via a single platform.

## Features :

- Multi tenant application support
- Device management
- Device notifications
- Device grouping
- Configuration management
- Reliable data delivery
- Data processing
- Secure communication
- Scalability, elasticity & self-healing



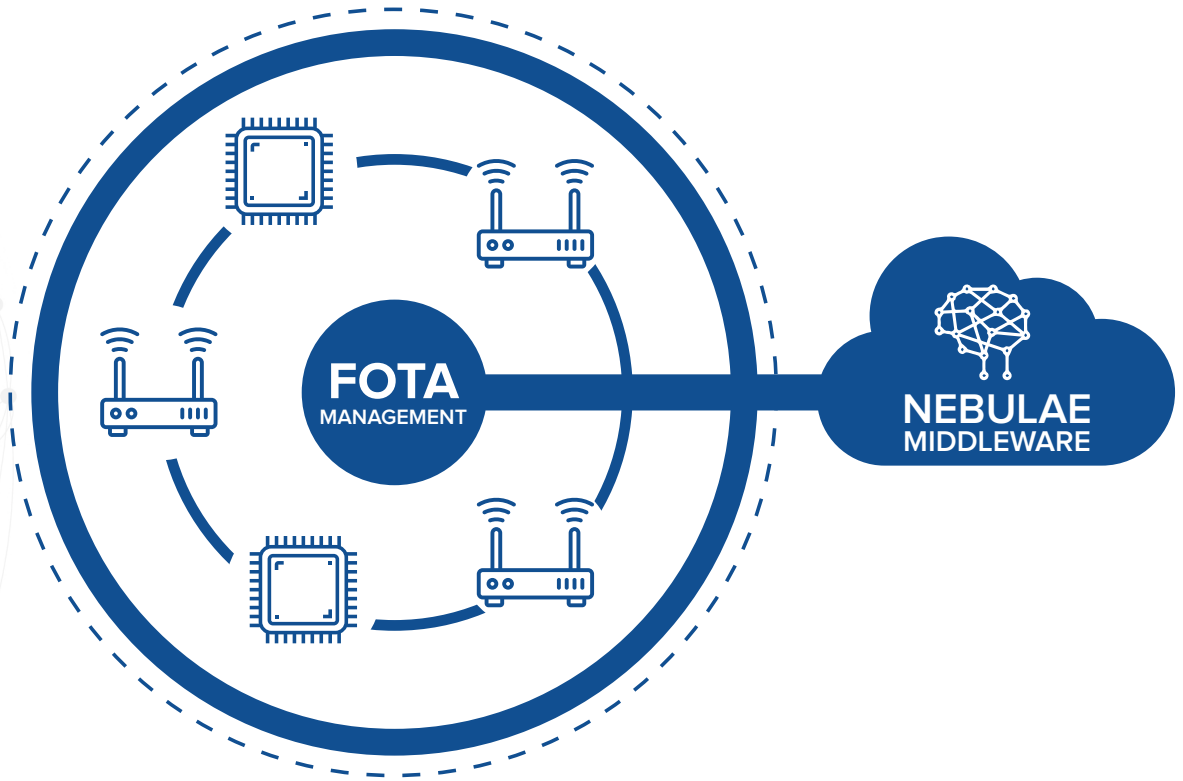
# NEBULINK FOTA

NebuLink FOTA is a key service of Nebulae to **protect IoT eco-system** from obsolescence, hacking and bugs. It can be used or available as a service into NebuLink Middleware or any 3rd party Middleware

Running on Node and Gateway, NebuLink FOTA service allow zero configuration with the shortest downtime with a single reboot on firmware updates.

## Features :

- 3 Layer verification:
- Integrity, authenticity, sanity
- A robust, reliable and fail-safe mechanism
- Group firmware update for similar devices
- Configurable chunk and bandwidth support
- Supports 2 layer device library support
- Firmware Update via Multicast and Broadcast Support
- Firmware utility to generate accurate FOTA images as per specification



# USE CASES

- SMART CITY
- SMART AGRICULTURE
- SMART MANUFACTURING



# SMART CITY

Nebulae smoothenes the smart city development by **integrating** the smart things network & enabling them to achieve **location**, intelligent recognition, **monitoring**, tracking and city **management**.



Smart Health



Security and Public Safety



Smart Parking



Smart Education



Smart Grid

# SMART AGRICULTURE

Nebulae aims at providing farmers with groundbreaking IoT solutions to **increase** their **production quality** with minimal efforts. **Placing sensors** and pinning down their values can be used to mimic the environmental conditions required by plants. Thereby **overcoming** the element of **poor weather** conditions and **human error**.



Pre-warn for  
potential crop diseases



Increase in  
crop yield



Remote monitoring  
of farms



Sensing wheather  
conditions



Data Analysis  
(Geo-location, weather, etc)

# SMART MANUFACTURING

Nebulae integrates technologies and protocols for networking, embedded control, sensing and **easy manageability** on which application software can run. The Nebulae IoT solution provides a platform to let industries to **remotely monitor** and **control Appliances/Machines**.



Automation of machines



Remote access & Control of sensors



Monitoring air quality level

# NOTABLE BREAKTHROUGHS





## SLS Divisions pioneering with the help of Nebulae IoT Platform



# LET'S GET CONNECTED



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# Thanks